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Ultrasonic device

Abstract

An ultrasonic sensor includes: a first substrate including an ultrasonic element; a first electrode and a second electrode on the first substrate; a second substrate having a through-hole that penetrates from a first surface facing the first substrate to a second surface; and a gap material that separates the first substrate and the second substrate from each other, in which in a plan view from a +Z direction, the through-hole overlaps with the first electrode and the second electrode, and the gap material surrounds the through-hole, the through-hole has a narrow portion, and a width of the narrow portion is smaller than a width of the through-hole in the first surface, in a direction orthogonal to the +Z direction.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7342351	12/2007	Kubo et al.	N/A	N/A
10296135	12/2018	Imazeki	N/A	H01L 23/3192
2010/0052473	12/2009	Kimura	29/25.35	H03H 9/059
2010/0270891	12/2009	Kohda	29/25.35	H03H 9/1035
2017/0031024	12/2016	Kiyose et al.	N/A	N/A
2017/0155028	12/2016	Kiyose	N/A	H10N 30/802
2021/0223213	12/2020	Matsuda	N/A	B06B 1/0662

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2010103479	12/2009	JP	N/A
2017029270	12/2016	JP	N/A
2004105237	12/2003	WO	N/A

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2020-155377, filed Sep. 16, 2020, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to an ultrasonic device.

2. Related Art

(3) In the related art, an ultrasonic device including ultrasonic elements has been known. In such an ultrasonic device, a plurality of elements and a plurality of terminal portions are provided on an element substrate, and the plurality of terminal portions and electrodes such as flexible printed circuits (FPCs) are electrically coupled to each other. For example, JP-A-2017-29270 discloses a wiring structure of an ultrasonic device in which a terminal portion of an element substrate and electrodes of FPCs are electrically coupled to each other via a through-hole provided in a sealing plate for protecting a plurality of elements.

(4) However, the wiring structure described in JP-A-2017-29270 has a problem that the manufacturing cost tends to increase. Specifically, the FPC is bent and inserted into the through-hole in a terminal region, and a contact is covered with a protective member. Therefore, a mounting process becomes complicated and is thus difficult to be automatized. In addition, since the FPC is mounted via the through-hole, the wiring structure also has a problem that the FPC is easily removed from the through-hole when a force such as a tensile force is applied to the FPC. That is, there has been a demand for an ultrasonic device capable of reducing manufacturing costs and implementing steady mounting with a simple configuration.

SUMMARY

(5) An ultrasonic device includes: a first substrate including an ultrasonic element; an electrode on the first substrate; a second substrate having a through-hole that penetrates from a first surface facing the first substrate to a second surface opposite to the first surface; and a gap material interposed between the electrode and the second substrate and separating the first substrate and the second substrate, in which the second substrate is stacked in a first direction with respect to the first substrate, in a plan view from the first direction, the through-hole overlaps with the electrode, and the gap material surrounds the through-hole, the through-hole has a narrow portion, and a width of the narrow portion is smaller than a width of the through-hole in the first surface, in a second direction orthogonal to the first direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic view illustrating a function of an ultrasonic sensor as an ultrasonic device according to a first embodiment.

(2) FIG. **2** is a schematic cross-sectional view illustrating a configuration around an ultrasonic element.

(3) FIG. **3** is a schematic plan view illustrating a form of a second substrate.

(4) FIG. **4** is a schematic cross-sectional view illustrating a form of the second substrate.

(5) FIG. **5** is a schematic cross-sectional view illustrating a form of a second substrate according to a second embodiment.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(6) In each of the following drawings, as necessary, X, Y, Z axes are assigned as mutually orthogonal coordinate axes, a direction indicated by each arrow is denoted as + direction, and a direction opposite to the + direction is denoted as a − direction. A +Z direction may be referred to as an upper side and a −Z direction may be referred to as a lower side. In the present specification, a first direction is referred to as the +Z direction. A second direction orthogonal to the first direction is a direction orthogonal to a direction along a Z axis, for example, a direction along an X axis.

(7) Furthermore, in the following description, for example, a description for a substrate of “on the substrate” indicates any one of a case in which a component is disposed on the substrate in contact therewith, a case in which a component is disposed on the substrate via another structure, or a case in which a part of a component is disposed on the substrate in contact therewith, and another part is disposed on the substrate via another structure.

1. First Embodiment

(8) In the present embodiment, an ultrasonic sensor is exemplified as an ultrasonic device having one or more ultrasonic elements that generate ultrasonic waves by vibration. A configuration of an ultrasonic sensor **1** according to a first embodiment will be described with reference to FIGS. **1** and **2**.

(9) As illustrated in FIG. **1**, the ultrasonic sensor **1** includes an ultrasonic transceiver **100** and a

timer **200**. The transceiver **100** transmits ultrasonic waves in a transmission direction D**1** and receives the ultrasonic waves reflected from an object O in a reception direction D**2**. The ultrasonic waves transmitted by the transceiver **100** are generated by an ultrasonic element on a transmission side, which will be described later. In addition, the ultrasonic waves received by the transceiver **100** are received by an ultrasonic element on a reception side, which will be described later.

(10) The timer **200** measures a time from when the transceiver **100** transmits ultrasonic waves to when the ultrasonic waves reflected from the object O are received. As a result, the ultrasonic sensor **1** measures a distance Lo between the ultrasonic sensor **1** and the object O.

(11) As illustrated in FIG. **2**, the transceiver **100** includes a first substrate **110** including an ultrasonic element **113**, a first electrode **111** and a second electrode **112** as electrodes, gap materials **114**, and a second substrate **115**. The ultrasonic element **113** includes the ultrasonic element **113** on the transmission side and the ultrasonic element **113** on the reception side. Since the ultrasonic element **113** on the transmission side and the ultrasonic element **113** on the reception side have the same configuration, and may be collectively simply referred to as the ultrasonic element **113** hereafter.

(12) The transceiver **100** has a vibration plate **110***a*, the first electrode **111** or the second electrode **112**, the gap material **114**, and the second substrate **115** which are stacked in this order from the first substrate **110** upward. The second substrate **115** is stacked in the +Z direction with respect to the first substrate **110** via the gap material **114** or the like.

(13) An opening **110***b* is formed in the first substrate **110**. The opening **110***b* penetrates the first substrate **110**. A vibration plate **110***a* is stacked on the first substrate **110**. Therefore, the vibration plate **110***a* in a region corresponding to the opening **110***b* functions as a vibratable film exposed in the −Z direction.

(14) When a voltage is applied to the first electrode **111**, the ultrasonic element **113** on the transmission side expands and contracts in a direction along the X axis and a direction along the Y axis. As a result, the vibration plate **110***a* in the region vibrates to generate ultrasonic waves. The ultrasonic waves are transmitted from the opening **110***b* corresponding to the ultrasonic element **113** on the transmission side toward the object O. After the ultrasonic waves are reflected from the object O, the vibration plate **110***a* vibrates via the opening **110***b*, and the vibration is received by being transmitted to the ultrasonic element **113** on the reception side. Therefore, the opening **110***b* is made to face the object O when using the ultrasonic sensor **1**.

(15) The materials of the first substrate **110** include, for example, silicon (Si), magnesium oxide (MgO), lanthanum aluminate (LaAlO.sub.3), sapphire, gallium arsenide (GaAs), zirconium oxide (ZrO.sub.2), and aluminum oxide (Al.sub.2O.sub.3), and the like are applied. Examples of a material of the vibration plate **110***a* can include silicon oxide (SiO.sub.2), silicon nitride (SiN), zirconium oxide (ZrO.sub.2), aluminum oxide (Al.sub.2O.sub.3), titanium oxide (TIO.sub.2), magnesium oxide (MgO), and lanthanum aluminate (LaAlO.sub.3), and hafnium oxide (HfO.sub.2).

(16) The first electrode **111** or the second electrode **112** is disposed on the first substrate **110** via the vibration plate **110***a*. The ultrasonic element **113** is disposed above the opening **110***b*. A part of a lower side of the ultrasonic element **113** is disposed to be in contact with the vibration plate **110***a*, and the other part thereof is disposed to be in contact with the first electrode **111**. A side surface and an upper side of the ultrasonic element **113** in the −X direction are in contact with the second electrode **112**. That is, one end of the ultrasonic element **113** and the first electrode **111**, and the other end of the ultrasonic element **113** and the second electrode **112** are electrically coupled to each other. Since the ultrasonic element **113** is electrically coupled to the first electrode **111** and the second electrode **112**, it is possible to drive the ultrasonic element **113** to generate ultrasonic waves or receive the ultrasonic waves.

(17) The gap material **114** is disposed in the −X direction and the +X direction, which are lateral to the ultrasonic element **113**, and the second substrate **115**, which is a protective substrate of the

ultrasonic element **113**, is disposed above the ultrasonic element **113**. The ultrasonic element **113** is housed in a space substantially enclosed vertically and laterally. The gap material **114** surrounds a through-hole **115***a* in a frame shape in a plan view from the +Z direction. The detailed form of the gap material **114** will be described later.

(18) A known piezoelectric body is applied to the ultrasonic element **113**. An Example of a material of the piezoelectric body can include a composite oxide having a perovskite (ABO.sub.3) structure. Specific examples thereof can include a lead-based composite oxide such as lead zirconate titanate (PZT), a bismuth iron acid (BFO)-based material, and a lead-free composite oxide such as sodium potassium niobate (KNN). Using the lead-based composite oxide becomes easier to secure a displacement amount of vibration in the ultrasonic element **113**. Using the lead-free composite oxide becomes easier to promote environmental friendliness.

(19) In the BFO-based material, bismuth (Bi) is located at A site of the perovskite structure and iron (Fe) is located at B site. Elements other than Bi, Fe, and oxygen (O) may be added to the BFO-based material. Examples of the elements can include manganese (Mn), aluminum (Al), lanthanum (La), barium (Ba), titanium (Ti), cobalt (Co), cerium (Ce), samarium (Sm), chromium (Cr), potassium (K), lithium (Li), calcium (Ca), strontium (Sr), vanadium (V), niobium (Nb), tantalum (Ta), molybdenum (Mo), tungsten (W), nickel (Ni), zinc (Zn), praseodymium (Pr), neodymium (Nd), and europium (Eu). Among them, one or more of the elements may be included.

(20) Elements other than K, sodium (Na), Nb, and O may be added to the KNN-based material. Examples of the elements can include Mn, Li, Ba, Ca, Sr, zirconium (Zr), Ti, Bi, Ta, antimony (Sb), Fe, Co, silver (Ag), magnesium (Mg), Zn, copper (Cu), V, Cr, Mo, W, Ni, Al, silicon (Si), La, Ce, Pr, Nd, promethium (Pm), Sm, and Eu. Among them, one or more of the elements may be included.

(21) The composite oxide having a perovskite structure includes those having a composition shifted from stoichiometric composition due to deficiency of atoms or presence of excess atoms in a crystal structure, and those whose element is partially substituted with other elements. That is, those having an inevitable shift of the composition due to lattice mismatch, oxygen deficiency, and the like, and partial substitution with the elements are allowed as long as the perovskite structure can be obtained.

(22) The materials of the first electrode **111** and the second electrode **112** are not particularly limited as long as they have conductivity. Examples of the materials of the first electrode **111** and the second electrode **112** can include metallic materials such as platinum (Pt), iridium (Ir), gold (Au), Al, Cu, Ti, and stainless steel, tin oxide-based conductive materials such as indium tin oxide (ITO) and fluorine-doped tin oxide (FTC)), a zinc oxide-based conductive material, oxide conductive materials such as ruthenium-based strontium, lanthanum nickelate, and element-doped strontium titanate, a conductive polymer, and the like.

(23) The gap material **114** is disposed on the first electrode **111** and the second electrode **112**. The gap material **114** is interposed between the first electrode **111** and the second substrate **115** and the second electrode **112** and the second substrate **115**. The first substrate **110** and the second substrate **115** are separated by the gap material **114** to form the above-described space in which the ultrasonic element **113** is housed.

(24) As will be described in detail later, the gap material **114** is located so as not to overlap with the ultrasonic element **113** and the opening **110***b* in a plan view from the +Z direction, and is formed so that each of the first electrode **111** and the second electrode **112** are individually partially surrounded in a frame shape. In the first electrode **111** and the second electrode **112**, a region surrounded by the gap material **114** in a frame shape serves as an electrical contact with a conductive resin as described later. That is, in each of the first electrode **111** and the second electrode **112**, the region surrounded by the frame-shaped gap material **114** becomes an electrode terminal that is simple and easy to be miniaturized. The detailed form of the gap material **114** will be described later.

(25) The gap material **114** is formed of, for example, a curable resin such as a photocurable type

resin. As a result, a distance between the first substrate **110** and the second substrate **115** in the direction along the Z axis is formed with high accuracy.
(26) The second substrate **115** is supported by the gap material **114** above the ultrasonic element **113**, the first electrode **111**, and the second electrode **112**. The second substrate **115** has a first surface **115***s***1** facing the first substrate **110** in a direction along the Z axis, and a second surface **115***s***2** opposite to the first surface **115***s***1**. The same material as that of the first substrate **110** is applied to the second substrate **115**.
(27) The second substrate **115** has the through-hole **115***a*. The through-hole **115***a* penetrates from the first surface **115***s***1** to the second surface **115***s***2** in the second substrate **115**. The through-hole **115***a* is disposed corresponding to each of the frame-shaped gap materials **114**, overlaps with the first electrode **111** and the second electrode **112** and does not overlap with the ultrasonic element **113**, in a plan view from the +Z direction. Since the ultrasonic element **113** is disposed in a space formed by the first substrate **110**, the second substrate **115**, and the gap material **114**, vibrations transmitted or received by the ultrasonic element **113** are not hindered.
(28) A recess **119** whose upper portion is opened by a region surrounded by the gap material **114** in a frame shape and the through-hole **115***a* is formed. The recess **119** is filled with a conductive resin **118**. The conductive resin **118** is electrically coupled to each of the first electrode **111** and the second electrode **112**. The conductive resin **118** serves as a connection wire for mounting in the transceiver **100**.
(29) The conductive resin **118** is formed by filling the recess **119** with a silver paste or the like and solidifying the silver paste or the like. An end on an upper portion of the conductive resin **118** is protruded upward from the second substrate **115** by making a volume of the filled conductive resin **118** larger than an internal volume of the recess **119**. The portion protruded upwardly in the conductive resin **118** serves as a small mounting terminal that is easy to be mounted.
(30) The detailed form of the gap material **114** will be described with reference to FIGS. **3** and **4**. In FIGS. **3** and **4**, for convenience of illustration, the ultrasonic element **113**, the vibration plate **110***a*, the conductive resin **118**, and the like are omitted, and other configurations are simplified. Further, in FIG. **3**, the gap material **114**, the first electrode **111**, the second electrode **112**, and the like shielded by the second substrate **115** are transmitted through the second substrate **115** and indicated by a broken line.
(31) As illustrated in FIG. **3**, in a plan view from the +Z direction, the through-hole **115***a* corresponding to the first electrode **111** overlaps with the first electrode **111**, and the through-hole **115***a* corresponding to the second electrode **112** overlaps with the second electrode **112**. That is, one first electrode **111** or one second electrode **112** corresponds to one through-hole **115***a*. Although not illustrated, the +Y direction of the first electrode **111** and the second electrode **112** leads to the wire on the first substrate **110**.
(32) The through-hole **115***a* has a narrow portion **117**. The narrow portion **117** is formed on the second surface **115***s***2**. That is, the narrow portion **117** is disposed at a boundary with the second surface **115***s***2** in the through-hole **115***a*. The narrow portion **117** is not limited to being disposed on the second surface **115***s***2** in the through-hole **115***a*. The through-hole **115***a* corresponding to the first electrode **111** and the through-hole **115***a* corresponding to the second electrode **112** have the same form. The details of the narrow portion **117** will be described later.
(33) The gap material **114** is formed in a frame shape and surrounds the through-hole **115***a* in a plan view from the +Z direction. The gap material **114** formed on the first electrode **111** and the gap material **114** formed on the second electrode **112** have the same forms.
(34) Since the gap material **114** is formed in a flat frame shape, it is possible to prevent the conductive resin **118** from flowing out and coming into contact with the ultrasonic element **113** or the like when the recess **119** is filled with the conductive resin **118**.
(35) As illustrated in FIG. **4**, the narrow portion **117** is in the through-hole **115***a* and is disposed on the second surface **115***s***2**. A width W**2** of the narrow portion **117** is smaller than a width W**1** of the

through-hole **115***a* in the first surface **115***s***1** in the direction along the X axis orthogonal to the +Z direction. That is, the width of the through-hole **115***a* becomes smaller in a tapered shape from the first surface **115***s***1** to the second surface **115***s***2**, in the direction along the X axis. Although the width W**1** and the width W**2** are illustrated in FIG. **4** when viewed from the −Y direction, a width relationship between the width W**1** and the width W**2** is not limited to a case where the width W**1** and the width W**2** are viewed from the −Y direction. The width W**1** and the width W**2** may correspond to a direction other than the direction along the X axis, which is orthogonal to the +Z direction.
(36) The above-described tapered shape of the through-hole **115***a* can be formed, for example, by performing wet etching on the second substrate **115** from the first surface **115***s***1** side. Accordingly, the narrow portion **117** can be easily formed.
(37) The recess **119** has a shape in which the first electrode **111** or second electrode **112** side is large and an upper portion is small in the through-hole **115***a*. Therefore, forming the conductive resin **118** in the recess **119** can make the conductive resin **118** difficult to be removed from the recess **119**.
(38) Shapes of the gap material **114** and the through-hole **115***a* are not limited to being substantially rectangular, and may be circular or elliptical, in a plan view from the +Z direction. In addition, a configuration in which each of the first electrode **111** and the second electrode **112**, and the second substrate **115** are in contact with the gap material **114** has been illustrated in the present embodiment, but the present disclosure is not limited thereto. The gap material **114** may be in indirect contact with the first electrode **111** and the second electrode **112**, or the second substrate **115** via an adhesive or the like.
(39) In the present embodiment, the ultrasonic sensor **1** that measures the distance Lo from the object O is illustrated as an ultrasonic device, but the ultrasonic device is not limited thereto. The ultrasonic device may be, for example, a flow rate sensor, an object detection sensor, an image sensor, a power generation element, or the like.
(40) According to the present embodiment, the following effects can be obtained.
(41) With a simple configuration, manufacturing costs can be reduced and mounting can be steadily implemented. Specifically, the first electrode **111** and the second electrode **112** are each surrounded by the frame-shaped gap material **114** in a plan view from the +Z direction, and one of the first electrode **111** and the second electrode **112** corresponds to one through-hole **115***a*. Therefore, the conductive resin **118** is filled from the through-hole **115***a* to the recess **119** to form a mounting terminal. Further, it is possible to form a mounting terminal by pulling the wire electrically coupled to the first electrode **111** and the second electrode **112** out from the through-hole **115***a* and filling a non-conductive material or the like in the recess **119**. As a result, an electrode terminal for mounting is formed with a simple configuration. Therefore, the work of bending an FPC becomes unnecessary, and the mounting process becomes easy to be automatized. In addition, an FPC insertion mechanism and a fixing mechanism are not required.
(42) Moreover, the through-hole **115***a* has a width that becomes smaller from the first surface **115***s***1** on the first substrate **110** side toward the narrow portion **117** at an upper side. Therefore, when the conductive material or non-conductive material flows into the through-hole **115***a* to form an electrode terminal in the mounting process, the narrow portion **117** functions to prevent the electrode terminal from being removed. Furthermore, as compared with the through-hole **115***a* formed of the surfaces along the Z axis, an adhesive area of the through-hole **115***a* and the conductive material or the like is increased, thereby improving an adhesive force. As a result, the electrode terminal can be steadily mounted. As described above, it is possible to provide the ultrasonic sensor **1** capable of reducing manufacturing costs and implementing steady mounting with a simple configuration.
2. Second Embodiment
(43) An ultrasonic sensor in the present embodiment is exemplified as an ultrasonic device as in the

first embodiment. The ultrasonic sensor according to the present embodiment has a different form of the through-hole **115***a* in the second substrate **115**, as compared with the transceiver **100** included in the ultrasonic sensor **1** of the first embodiment. Therefore, the same reference numerals are used for the same constituent parts as those in the first embodiment and duplicate descriptions thereof will be omitted.
(44) A detailed form of the second substrate **125** in a transceiver **220** according to the present embodiment will be described with reference to FIG. **5**. In FIG. **5**, the ultrasonic element **113**, the vibration plate **110***a*, the conductive resin **118**, and the like are omitted for convenience of illustration, and other configurations are simplified, as in FIG. **4** of the first embodiment.
(45) As illustrated in FIG. **5**, the transceiver **220** included in the ultrasonic sensor of the present embodiment includes a first substrate **110** including an ultrasonic element **113** (not illustrated), a first electrode **111** and a second electrode **112** on the first substrate **110**, a second substrate **125** having a through-hole **125***a*, and gap materials **114** interposed between the first electrode **111** and the second substrate **125** and between the second electrode **112** and the second substrate **125** and separating the first substrate **110** and the second substrate **125**.
(46) The second substrate **125** is supported by the gap material **114** above the ultrasonic element **113**, the first electrode **111**, and the second electrode **112**. The second substrate **125** has a first surface **125***s***1** facing the first substrate **110** in a direction along the Z axis, and a second surface **125***s***2** opposite to the first surface **125***s***1**. The same material as that of the second substrate **115** of the first embodiment is applied to the second substrate **125**.
(47) The second substrate **125** has the through-hole **125***a*. The through-hole **125***a* penetrates from the first surface **125***s***1** to the second surface **125***s***2** in the second substrate **125**. The through-hole **125***a* is disposed corresponding to each of the frame-shaped gap materials **114**, overlaps with the first electrode **111** and the second electrode **112** and does not overlap with the ultrasonic element **113**, in a plan view from the +Z direction. Since the ultrasonic element **113** is disposed in a space formed by the first substrate **110**, the second substrate **125**, and the gap material **114**, vibrations transmitted or received by the ultrasonic element **113** are not hindered.
(48) A recess **129** whose upper portion is opened by a region surrounded by the gap material **114** in a frame shape and the through-hole **125***a* is formed. The recess **129** is filled with a conductive resin **118** (not illustrated). The conductive resin **118** is electrically coupled to each of the first electrode **111** and the second electrode **112**. The conductive resin **118** serves as a connection wire for mounting in the transceiver **100**.
(49) A narrow portion **127** is in the through-hole **125***a* and is disposed between the first surface **125***s***1** and the second surface **125***s***2**. A width W**4** of the narrow portion **127** is smaller than a width W**3** of the through-hole **125***a* in the first surface **125***s***1** and a width W**5** of the through-hole **125***a* in the second surface **125***s***2**, in the direction along the X axis orthogonal to the +Z direction. That is, the through-hole **125***a* becomes smaller in a tapered shape from the first surface **125***s***1** toward the narrow portion **127**, and becomes smaller in a tapered shape from the second surface **125***s***2** toward the narrow portion **127**. The narrow portion **127** is provided in a form projecting from the recess **129**. In this case, although the widths W**3**, W**4**, and W**5** are illustrated in FIG. **5** when viewed from the −Y direction, a width relationship between the widths W**3** and W**5** and the width W**4** is not limited to a case where the widths W**3**, W**4**, and W**5** are viewed from the −Y direction. The widths W**3**, W**4**, and W**5** may correspond to a direction other than the direction along the X axis, which is orthogonal to the +Z direction.
(50) The narrow portion **127** of the through-hole **125***a* can be formed by, for example, performing wet etching on the second substrate **125** from two directions, the first surface **125***s***1** side and the second surface **125***s***2** side. Accordingly, the narrow portion **127** can be easily formed.
(51) As a result, the recess **129** has a shape in which the middle is small and upper and lower portions are large in the direction along the Z axis in the through-hole **125***a*. Therefore, forming the conductive resin **118** in the recess **129** can make the conductive resin **118** difficult to be removed

from the recess **129**.
(52) The width relationship between the width W**3** and the width W**5** is not particularly limited, but it is preferable that the width W**5** is smaller than the width W**3**. Accordingly, in addition to the narrow portion **127**, the through-hole **125***a* on the second surface **125***s***2** side is smaller than the through-hole **125***a* on the first surface **125***s***1** side, and thus, the mounting terminal formed in the recess **129** can be made more difficult to remove.
(53) Shapes of the gap material **114** and the through-hole **125***a* are not limited to being substantially rectangular, and may be circular or elliptical, in a plan view from the +Z direction. In addition, a configuration in which the second substrate **125** is in contact with the gap material **114** has been illustrated in the present embodiment, but the present disclosure is not limited thereto. The gap material **114** may be in indirect contact with the second substrate **125** via an adhesive or the like.
(54) In the present embodiment, the ultrasonic sensor that measures the distance Lo from the object O is illustrated as an ultrasonic device, but the ultrasonic device is not limited thereto.
(55) According to the present embodiment, the same effect as that of the first embodiment can be obtained.

Claims

1. An ultrasonic device comprising: a first substrate including an ultrasonic element; an electrode on the first substrate; a second substrate having a through-hole that penetrates from a first surface facing the first substrate to a second surface opposite to the first surface; a gap material formed of a curable resin and interposed between the electrode and the second substrate and separating the first substrate and the second substrate; and a conductive resin electrically coupled to the electrode and disposed on a space surrounded by the electrode, the through-hole, and the gap material, wherein the gap material is in direct contact with the second substrate, the second substrate is stacked in a first direction with respect to the first substrate, the conductive resin protrudes from the second substrate toward the first direction, in a plan view from the first direction, the through-hole overlaps with the electrode, and the gap material surrounds the through-hole, the through-hole has a narrow portion, a width of the narrow portion is smaller than a width of the through-hole in the first surface, in a second direction orthogonal to the first direction, and a width of the space surrounded by the electrode, the through-hole, and the gap material is equal to the width of the through-hole in the first surface.
2. The ultrasonic device according to claim 1, wherein the narrow portion is formed at the second surface, and the width of the through-hole becomes smaller from the first surface toward the second surface in the second direction.
3. The ultrasonic device according to claim 1, wherein the narrow portion is formed between the first surface and the second surface.
4. The ultrasonic device according to claim 1, wherein the ultrasonic element does not overlap with the through-hole in the plan view from the first direction.
5. The ultrasonic device according to claim 2, wherein the ultrasonic element does not overlap with the through-hole in the plan view from the first direction.
6. The ultrasonic device according to claim 3, wherein the ultrasonic element does not overlap with the through-hole in the plan view from the first direction.
7. The ultrasonic device according to claim 1, wherein the narrow portion is formed at a middle portion of the first surface and the second surface, the through-hole becomes smaller in a tapered shape from the first surface to the middle portion, and the through-hole becomes smaller in the tapered shape from the second surface to the middle portion.
8. The ultrasonic device according to claim 1, wherein a volume of the conductive resin is larger than an internal volume of the space surrounded by the electrode, the through-hole, and the gap

material.

9. The ultrasonic device according to claim 1, wherein at least a portion of the electrode is on the ultrasonic element, and at least the portion of the electrode is in direct contact with the ultrasonic element.